

Primewave AppV1

What loads when the app starts

Cold-start sequence, functions, SecureStore keys, Home Assistant WebSocket commands, and admin APIs — from native splash to home screen.

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Entry path: Expo Router → `app/_layout.jsx` → `app/index.jsx` (Lottie splash) → either `/login` or `/dashboard-v2`. Logged-in users land on `dashboard-v2`. The home screen is shown only after HA entity states plus quick-scenes and home-access APIs have returned.

1. Startup sequence (high level)

These phases overlap. Fonts, push registration, and icon preload start in the root layout while the splash animation plays. Dashboard then fires Home Assistant, admin mappings, Frigate, notifications, and heartbeat in parallel.

Phase	Where	Blocks UI?	Outcome
0. Native shell	Expo / iOS / Android	Yes — OS splash	JS bundle starts. Native splash stays until fonts + Lottie hide it.
1. Root layout	app/_layout.jsx	Yes — until fonts	Fonts, crash handler, push, orientation, light-icon preload, notification tap capture.
2. Splash route	app/index.jsx	Yes — min 2.2s	Lottie PrimeWave2.json. Session check in SecureStore. Then login or dashboard.
3. Session gate	index.jsx → router.replace	No extra wait	Valid session → /dashboard-v2. Otherwise → /login.
4. Dashboard boot	app/dashboard-v2.jsx	Skeletons until ready	Profile, HA WebSocket, registries, admin APIs, Frigate, notifications, heartbeat.
5. Home reveal	revealStep → 5	Home hidden until gate	Shown when entities.length > 0 AND quick-scenes AND home-access have finished.

Tablet vs phone is decided later with `useDeviceType()` (short side ≥ 768). Both use dashboard-v2; layout columns change, it is not a separate load path.

2. Phase 1 — Root layout (app/_layout.jsx)

This file wraps every screen. It runs once per process, before splash or dashboard.

Function / API	File	What it does on load
<code>SplashScreen.preventAutoHideAsync()</code>	expo-splash-screen	Keeps the native splash visible until fonts are ready.
<code>useFonts({ ClashDisplay-* })</code>	expo-font / _layout.jsx	Loads 6 Clash Display OTF weights. Until this finishes, <code>RootLayout</code> returns null.
<code>preloadLocalLightIcons()</code>	utils/lightTypeAssets.js	Reads bundled light SVGs (up lights, spot, chandelier, track, coved LED, generic bulb) and tints them for white/black.
<code>registerForPushNotificationsAsync()</code>	services/notifications.js	Android channel, permission prompt, Expo push token, POST <code>/api/notifications/register</code> .
<code>ScreenOrientation.lockAsync(PORTRAIT_UP)</code>	expo-screen-orientation	Phones only (short side < 768). Tablets stay free to rotate.
<code>Notifications.getLastNotificationResponseAsync()</code>	expo-notifications	Cold-start: if the user launched the app by tapping a push, capture title/body/category into <code>NotifContext</code> .
<code>addNotificationResponseReceivedListener()</code>	expo-notifications	Later taps while running/suspended also go into <code>NotifContext</code> .
<code>ErrorUtils.setGlobalHandler()</code>	_layout.jsx	Logs fatal/non-fatal JS crashes before the default handler.
<code>SplashScreen.hideAsync()</code>	_layout onLayout	Hides native splash once <code>fontsLoaded</code> is true.

Also mounted here: **NotifContext.Provider**, **ErrorBoundary**, **GestureHandlerRootView**, Expo Router **Stack** (index, login, dashboard-v2, room, ...), and on iOS a non-interactive purple shadow overlay.

3. Phase 2 — Splash screen (app/index.jsx)

Minimum display time is **2200 ms** so the Lottie animation can play even if `SecureStore` is instant.

Function / key	Purpose
LottieView source=PrimeWave2.json	Full-screen branded animation (loop, speed 0.6). Hides native splash on layout / onAnimationLoaded.
SecureStore.getItemAsync('is_logged_in')	Must be the string 'true'.
SecureStore.getItemAsync('ha_active_profile_id')	Which HA profile to use.
SecureStore.getItemAsync('ha_profiles')	JSON array of profiles (haUrl, haToken, adminUrl, name, userId).
SecureStore.getItemAsync('logged_in_user')	JSON { name, userId } passed as dashboard route params.
router.replace('/dashboard-v2')	If all four keys are valid and the active profile exists.
router.replace('/login')	Any missing key, parse error, or unknown profile id.

Login is skipped when those four SecureStore keys are present. Face ID / password are not re-checked on a normal warm or cold start with a saved session.

4. If there is no session — login (app/login.jsx)

Only this path runs when splash sends the user to /login.

- **checkBiometrics()** — expo-local-authentication: hardware + enrolled Face ID / fingerprint.
- **loadSettings()** — reads ha_profiles, ha_active_profile_id, Face ID flags, saved biometric creds.
- Optional **scanNetwork()** (utils/discovery.js) if the user scans for Home Assistant.
- On submit: **validateCredentials()** (services/auth.js), **upsertAccountAndActivate()**, then **registerForPushNotificationsAsync()** again, then navigate to dashboard-v2.

5. Phase 4 — Dashboard boot (app/dashboard-v2.jsx)

On first mount, dashboard-v2 starts several independent jobs. None of them wait for each other except the home-reveal gate in section 7.

5.1 Local settings (SecureStore, parallel)

Key	State it fills
settings_show_family	showFamily — person badges
settings_auto_room_visit	autoRoomVisit — open room from presence
settings_auto_room_resume	autoRoomResume — same on app resume
settings_show_voice_assistant	showVoiceAssistant — Butler mic in header
settings_show_preference_button	showPreferenceButton
room_reorder_config	savedRoomOrder — custom room card order
ha_active_profile_id + ha_profiles	connectionConfig { url, token, adminUrl, loaded }

Functions: `loadConnectionConfig()`, `getButlerBackendUrl()` (resolves adminUrl + /api/butler, no network yet).

5.2 Home Assistant WebSocket (HAService)

Once `connectionConfig.loaded` is true and `haUrl` + `haToken` exist: `new HAService(url, token) → connect() → WebSocket` to `/api/websocket`.

HA command	Function	Result in app
auth (access_token)	HAService.sendAuth()	auth_ok → status CONNECTED; else AUTH_FAILED
subscribe_events / state_changed	handleMessage after auth_ok	Live entity updates for the rest of the session
get_states	getStates()	Full entity list (lights, climate, covers, locks, sensors, ...)

HA command	Function	Result in app
get_config	getConfig()	location_name → cityName in the header
config/device_registry/list	getDeviceRegistry()	Devices → area mapping
config/entity_registry/list	getEntityRegistry()	Entity → device / area / labels
config/category_registry/list	getCategoryRegistry()	Categories
config/area_registry/list	getAreaRegistry()	Rooms / areas for the home grid
config/floor_registry/list	getFloorRegistry()	Floors; first floor selected by default
config_entries/get	getConfigEntries()	Music Assistant entry_ids (failure ignored)

Those eight registry/state calls run in **one Promise.all** after auth_ok, then React state is set in a single batch. HAService also listens to AppState and reconnects (up to 5 times, exponential backoff) when returning to foreground.

5.3 Admin backend APIs (AppBackendV1)

Fired from fetchMappings() and the same connection-config effect. All use Bearer HA token. Several endpoints are requested twice (mappings + initial-load effect); the later response wins.

API	Function / caller	Used for
GET /api/config	dashboard initial-load effect	badgeConfig (selected areas, locks_armed, still-open reminder, ...). Sets adminStatus OK/ERROR.
GET /api/quick-scenes	fetchMappings()	allowedQuickScenes (max 4 on home). Sets scenesFetched — part of reveal gate.
GET /api/monitored-entities?type=light	fetchEnrichedLightMappings()	Light mappings (type, dimmable/CCT/RGB).
GET /api/light-types	fetchEnrichedLightMappings()	Icon type catalog; then preloadLocalLightIcons() again.
GET /api/monitored-entities?type=media_player	fetchMappings()	TV / Apple TV / media mappings.
GET /api/sensors	fetchMappings() + initial-load	Door / window / motion sensor types for badges and rooms.
GET /api/covers	fetchMappings() + initial-load	coverType, garageDuration, cover windows.
GET /api/climate-mappings	fetchMappings()	AC ↔ damper mappings.
GET /api/home-access	fetchMappings()	Which locks and garage/shutter covers appear on Home Access. Sets homeAccessFetched — reveal gate.
GET /api/lock-passage	fetchMappings()	Lock ↔ door-sensor pairs for passage mode.
GET /api/alerts	initial-load effect	Custom alert rules (kept in alertRulesRef for the socket handler).
GET /api/entities	initial-load effect	Monitored vs ignored entities for in-app notifications.
GET /api/room-tracking/lookup	initial-load effect	ESPresence / tracker state → area_id for auto-open room.
GET /api/cameras	initial-load effect	HA camera fallback if Frigate config is empty.
GET /api/notifications/history	useNotifications()	Entity-history notification list.
GET /api/notifications/log	useNotifications()	Lock/garage push log from the backend.
POST /api/sessions/heartbeat	startHeartbeat()	Immediately, then every 30s. Device id, user, foreground/background.

API	Function / caller	Used for
POST /api/notifications/register	registerForPushNotificationsAsync() (from layout)	Expo push token + userId + device name.

5.4 Frigate cameras

- **new FrigateService(", null, null, adminUrl, haToken)** — traffic is proxied through the admin backend, not a raw Frigate URL on the phone.
- **frigateService.getConfig()** — camera names/objects. Sets frigateCameras. Always sets frigateConfigResolved (success or fail).
- Home UI does **not** wait for Frigate. The camera strip fills in when this returns; the rest of home can already be visible.

5.5 Derived data after HA states arrive (no extra network)

Computation	Source
weather, humidity, indoorTemp	First weather.* entity + humidity/temp sensors
roomsWithCounts / dashboard rooms	getSelectedAreasForDashboard, filterParentRoomsForDashboard, getRoomEntities
Home Access locks & garages	selectedLockIds / selectedCoverIds + live entity states
Status badges (lights / AC / doors)	countActiveCountableLights, climate on, door sensors
Security alert items	Open locks / garage / shutters while alarm is armed
Quick scene tiles	allowedQuickScenes ∩ live entities
useHaSystemHealth()	Banner if HA/admin down; warnings delayed 10s
Presence tracker entity	Match userName in entity_id (*_room / *_location)

6. Phase 5 — When the home screen actually appears

Until the gate passes, dashboard-v2 shows **DashboardSkeleton** (header, scenes, home access, rooms, cameras placeholders).

Condition	Flag / data	If missing
HA states loaded	entities.length > 0	revealStep stays 0 (skeletons)
Quick scenes API finished (ok or error)	scenesFetched === true	Wait — even an empty list counts as fetched
Home Access API finished (ok or error)	homeAccessFetched === true	Wait
User opened the app from a push	alertNotif set	Force revealStep = 5 immediately so the alert modal can show

Frigate, light mappings, media, covers, climate, heartbeat, and Butler are **not** on this gate. They populate as they arrive.

7. After first paint (still during “load”, not on tap)

- **AppState listener** — on resume: re-run presence room, fetchMappings({ resetRevealCascade: false }), update heartbeat appState.
- **navigateToPresenceRoom()** — if autoRoomVisit is on and the user tracker reports a room, open that RoomSheet.
- **SecurityAlertModal** — if alarm is armed and a lock/garage/shutter is open.
- **NotificationModal / AlertNotificationModal** — pendingNotif from root layout.
- **Butler** is *not* connected at startup. Voice WebSocket + HA sync run only when the user taps the mic (canOpenButlerCall → runButlerBackgroundSetup → preflightButlerCall + POST /api/butler/admin/save).
- AI / Brain tab is not mounted until first visit (aiTabVisited).
- Room detail page (app/room.jsx) loads only when a room is opened; bootstrap is prepared by setRoomPageBootstrap().

8. Bundled assets loaded at startup

Asset	When
Clash Display fonts (6 weights)	Root layout — blocks first frame
PrimeWave2.json (Lottie)	Splash route
shadow.png	Root layout overlay (iOS only)
Light type SVGs (6 files)	preloadLocalLightIcons() on layout mount + again after light-types API
app-icon.png / splash background #09091A	Native Expo splash (app.json)

9. Intentionally not loaded at startup

- Butler Gemini Live WebSocket and microphone (until mic tap).
- Room page HA entities beyond the home registries (room screen has its own view).
- Frigate event history / clips (Cameras tab / modal on demand).
- Statistics, insights, automations, history screens.
- TV remote / Apple TV modal data until opened.
- Adaptive lighting switch details until the room lights card needs them.
- Full notification history refresh loop (one shot at mount via useNotifications).
- Re-authentication / Face ID when a session already exists.

10. Key files in the load path

File	Role on startup
app/_layout.jsx	Fonts, splash hide, push, orientation, NotifContext, stack
app/index.jsx	Lottie splash + session gate
app/login.jsx	Only if no session
app/dashboard-v2.jsx	Main boot: HA + admin + Frigate + UI reveal
services/ha.js	WebSocket client (auth, subscribe, get_states, registries)
services/frigate.js	Frigate config via admin proxy
services/notifications.js	Push token + backend register
services/heartbeat.js	POST /api/sessions/heartbeat every 30s
hooks/useNotifications.js	GET history + log
hooks/useHaSystemHealth.js	Connection banner (10s debounce)
hooks/useDeviceType.js	Phone vs tablet layout
utils/lightMappingsClient.js	Lights + types + icon inference
utils/lightTypeAssets.js	SVG icon preload
utils/butlerBackend.js	Resolve Butler URL (no socket yet)
utils/roomHelpers.js / roomAreas.js	Build room cards from registries

Practical takeaway: a logged-in cold start is bound by (1) font load, (2) 2.2s splash minimum, (3) HA WebSocket auth + get_states/registries, and (4) /api/quick-scenes + /api/home-access. Everything else (cameras, mappings, heartbeat, push, Butler) is background and must not block the first home paint.